Satya S Vepada

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5 years of Progressively responsible experience as Data Engineer Data Engineering, Data Analysis, ETL, and Data warehousing, Business Intelligence background in Designing, Developing, Analysis, Implementation, and post implementation support of DWBI applications with a proven track record of designing and implementing robust and scalable data solutions. volume data using Hadoop/Spark frameworks for business intelligence/data warehousing and data science applications. Skilled in big data management, ETL workflows, and dimensional modeling.

- Expertise in ETL processes, data modeling, and database management, ensuring optimal data flow, organization, and retrieval.
- Proficient in utilizing cloud platforms, including AWS and Azure, to architect and implement end-to- end data pipelines for diverse business needs.
- Skilled in leveraging big data technologies such as Apache Spark and Hadoop to process and analyze large datasets efficiently.
- Experienced in implementing data quality and governance measures, ensuring data accuracy, compliance, and security.Demonstrated ability to collaborate with cross-functional teams to understand business requirements and translate them into effective data solutions.
- Strong background in optimizing database performance, conducting query tuning, and implementing best practices for efficient data storage and retrieval.
- Engineered data pipelines in Azure Data Factory (ADF), orchestrating ETL processes from diverse sources to Azure SQL, Blob storage, and Azure SQL Data Warehouse. Hands-on experience with SQL, NoSQL databases, and data warehouse solutions, contributing to seamless integration and accessibility of data.
- Proven expertise in version control systems (e.g., Git) and agile methodologies, ensuring a collaborative and efficient development process.
- Passionate about staying current with emerging data technologies and trends, continuously seeking opportunities to enhance data engineering capabilities.
- Worked on architecting, designing, and implementation of large-scale data and analytics solutions on Snowflake Cloud Data Warehouse.
- Efficient in preprocessing data including Data cleaning, Correlation analysis, Imputation, Visualization, Feature Scaling and Dimensionality Reduction techniques using Machine learning platforms like Python Data Science Packages(Scikit-Learn, Pandas, NumPy), analyzing data using Python, R and SQL.
- Experience in building reliable ETL processes & data pipelines for batch and real - time streaming using SQL, Python, Databricks, Spark, Streaming, flink, Sqoop, Hive, AWS, Azure, NiFi, Oozie and Kafka.
- Responsible for designing and building new data models and schemas using Python and SQL. Involved in developing python scripts, SSIS, Informatica, and other ETL tools for extraction, transformation, loading of data into the data warehouse.
- Building the Tableau dashboards to provide the effectiveness of weekly campaigns and customer acquisition.
- Utilized AWS EMR to transform and move large amounts of data into and out of other AWS Data stores and databases, such a S3 and DynamoDB. To support business intelligence objectives, I built data warehousing systems utilizing Amazon S3, Dynamo DB, EC2, and Snowflake.
- Implements best practices to create cloud functions, applications, and databases. Worked in all stages of SoftwareDevelopment Life Cycle (SDLC).
- Experience loading tables from the azure data lake to azure blob storage for pushing them to snowflake.

TECHNICAL SKILLS

Programming: Python, SQL,Scala, Java, Unix/Shell Scripting
Big Data Technologies: Hadoop, Spark, Kafka, Databricks, Cloudera Distribution Hadoop, OOZIE, Airflow
Cloud and Database: Azure Data Factory, HDInsight, VMs, Blob Storage, Amazon EC2, S3, RDS , PostgreSQL, MS SQL Server, Oracle SQL **Big Data Technologies**: Hadoop, MapReduce, Hive, Sqoop, Oozie, EMR, Spark, Pig, lambda functions, Zookeeper), kafka, confluent kafka, Frameworks & Tools: (NumPy, Scikit-learn, Pandas, Seaborn, Pyspark, Power BI, Tableau, Git, GitHub, MS Excel, MS Access, PySpark, Airflow, ETL, Jira, Informatica power center, Waterwall, Agile/Scrum, Jenkins, Docker, Confluence.
Datastores and Cloud: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), MySQL, Oracle, MongoDB, Teradata, PostgreSQL, Cassandra, Apache Kafka, Snowflake, Data Bricks
Cloud Computing: AWS (S3, CloudWatch, Athena, RedShift, EMR, EC2, DynamoDB), Azure (Azure Data Factory, Azure Blob, Azure

Databricks), IAM, Secret Manager, S3, Lambda, CloudWatch, Messaging Queue (SNS & SQS), Azure - ADF, Blob Storage, Blob Storage, Amazon EC2, S3, RDS , PostgreSQL, MS SQL Server, Oracle
IDE and Platforms: Pycharm, VS, Eclipse, IntelliJ, Jupyter Notebook, JIRA, Service Now, Github, Putty, Linux, Windows, Azure Devops, Jenkins, CI/CD

WORK EXPERIENCE

Application Programmer - Global business , Bank of America **10/2023 – Present | Plano, TX**

Architecting robust, scalable data pipelines and ETL processes Optimizing data ingestion, processing, and analysis workflows. Translating business requirements into actionable technical solutions ,Collaborating cross- functionally & communicating technical concepts effectively.

* Data Integration: Proficient in Sqoop for efficient relational to Hadoop ecosystem data ingestion.
* Big Data Analytics: Extensive experience with Teradata parallel DBMS for handling large structured datasets.
* Data Warehousing: Skilled in Hive for distributed querying and analysis of big data on HDFS. Job Scheduling: Expertise in Autosys for automated execution of complex data pipelines.
* Data Visualization: Adept at insightful dashboard and report creation using Tableau
* Continuously enhanced and refactored the codebase to meet evolving requirements, with a strong emphasis on handling Personally Identifiable Information (PII) data and ensuring compliance with the GDPR.
* Migrated existing Hive/SQL queries to Spark DataFrame-based transformations, enhancing data processing efficiency and scalability.
* Skillfully triaged, analyzed, and resolved production issues, ensuring system stability and optimal performance by tracking root cause analysis.
* Worked on reconciliation data from Teradata using Python, SQL. Played key role in Migrating Teradata objects into Snowflake environment
* Collaborate closely with the data science team, worked on datasets for data exploration and various experiments by leveraging PySpark, HiveQL, or SQL to extract data from the data lake. Experiments with different file formats, such as Parquet and JSON, as well as various compression techniques to store raw historical data

Senior Software Engineer - Data Engineering, Sify - NBME **07/2022- 08/2023 |Remote, PA**

Built from scratch the end-to-end migration of NBME's on-premise data infrastructure to a robust and scalable AWS architecture utilizing IaC best practices.

* Active role in POC testing and development of data integration, processing and analytics pipelines leveraging AWS services like Glue, Lambda, Kinesis, EMR, S3, Redshift
* Built Python, PySpark ETL scripts and Spark SQL jobs to transform PDF, XML, image and gif data into JSON, Parquet formats with partitioning for analytical workloads
* Developed Snowflake stages, streams, tasks, pipelines for streaming integration and loading of processed data from S3
* Implemented Snowflake secure data sharing and cross-account access with customer accounts for analytics
* Wrote Spark SQL jobs for transformation of semi-structured data into analytics-ready formats in Delta Lake. Built reusable Scala UDFs leveraging Spark SQL for custom parsing and validation of ingestion streams Implemented serverless ingestion workflows using Lambda functions for streaming and batch data intake from SFTP and internal systems via Dell Boomi
* Designed a multi-account AWS environment spanning dev, test, prod accounts secured via derived KMS keys and service roles for inter-account communication
* Automated CICD pipelines using CodePipeline, CodeCommit and CodeBuild for secure and efficient deployment across accounts and environments. Initially using step functions, lambda trigger & later designed a workflow with Airflow, DAGs
* Led definition and build of IAM policies, VPC design, security groups, and other access controls following least privilege and security best practices
* Drove documentation of architecture, infrastructure, pipelines, security and operational processes using JIRA, Confluence and diagrams.
* Experimented with different file formats like Parquet, JSON and compression types for analyzing efficient storage of raw historical data
* Enhanced and refactored codebase to meet new requirements while handling PII data and ensuring GDPR compliance
* Automated build and deployment of CI/CD pipelines using Shell and Python scripts invoked via CodeBuild Partnered with analytics team to design and develop Redshift schemas, Glue catalogs and QuickSight dashboards to deliver business insights

* Implemented logging, monitoring and alert using CloudWatch and CloudTrail for operational visibility and rapid issue resolution
* Mentored 2 junior engineers on AWS big data services, Spark/EMR, CI/CD, infrastructure as code and security best practices
* Performed a POC on GCP Dataproc, GCS, Cloud functions,. GCP, Big Query, GCS bucket, G - cloud function, cloud dataflow, Pub/suB cloud shell, GSUTIL, BQ command line utilities, Data Proc, Stack driver for cost optimization at the end.
* Become a Subject Matter Expert in Data Engineering with a focus on GCP native services and other well integrated third-party technologies

Data Engineer | AT&T | USA **SEP 2021 - MAY 2022**

* Evaluated and implemented new technologies and tools to enhance data engineering capabilities, staying abreast of industry best practices.
* Developed data wrangling through data gathering and querying process over SQL and implemented data cleaning process with R and Python.
* Worked with a diverse set of clients all over the world, identifying and analyzing Dynamic Values, and managing development with agile methodology. Worked in a Unix environment with Oracle SQL Developer, Microsoft SQL Developer, Oracle Warehouse, and PyCharm.
* Presented Dashboards to Management for more Insights using Microsoft Power BI. Queried data from SQL server, imported other formats of data and performed data checking, cleansing, manipulation, and reporting using SQL.
* We also maintained a MySQL database for transient transactional data that required low-latency querying at various stages. The reconciler app leveraged MySQL extensively.
* The reconciler was a custom Python app I developed for data quality checks and reprocessing failed transactions. It compared source, transform and output data to ensure integrity.
* Enhanced the SQL scripts and NoSQL databases including MySQL, PostgreSQL, MongoDB, and Cassandra.
* Created new data pipelines to ingest data and orchestrate end to end data processing in Azure Data Factory involving Metadata lookup, Databricks activity, copy activity, Web activity etc.
* Supported building and release pipelines on Azure DEVOPS and involved in the deployments across DEV, UAT and PROD, achieving a 25% improvement in the pipeline runtime.
* Implemented ETL processes using tools such as Apache NiFi, Talend, and custom Python scripts to ingest, transform, and load data from various sources. Orchestrated and managed data workflows using Apache Airflow, ensuring timely and reliable execution of data processing tasks.
* Proficient in Azure services including Data Factory, Databricks, and Cosmos DB. Leveraged Azure Data Factory to streamline the extraction and loading of JSON data from REST APIs, resulting in data accuracy within the Azure SQL Server environment.
* Ingested and processed real-time streaming data from AT&T and DirecTV customers using Azure Stream Analytics, enabling near real-time analysis and insights.
* Optimized data storage and querying by employing Azure Data Lake Store and Azure Data Lake Analytics for big data analytics workloads.
* Implemented Azure Synapse Analytics (formerly Azure SQL Data Warehouse) for enterprise data warehousing and advanced analytics on large-scale datasets.
* Leveraged Azure Data Catalog for data source discovery and documentation, improving data governance and self-service analytics capabilities.
* Supported the Production jobs across various platforms, ETL jobs on Hadoop cluster with critical SLAs &high visibility using YARN, MapReduce, Hive, Pig, Sqoop
* Developed applications to process viewership, click stream and account activity data of AT&T and Direct TV customers to build BI models, Analyzed applications to provide a comprehensive overview on the Confidential and AT&T Customers' Viewership.
* Automated batch data ingestion from RDBMS and S3 into HDFS using Sqoop and shell scripts Developed MapReduce jobs to transform semi-structured data into Parquet format Implemented Hive for data discovery on HDFS and Oozie for job orchestration
* Used custom settings and custom meta data to store institution specific data and embed CloudFormation in terraform.

Data Engineer | Magna Infotech | India **AUG 2018 - JUL 2019**

* Designed, developed, and maintained scalable/fault tolerant efficient data pipelines, reducing data processing time by 30%.
* Implemented data quality checks and validation processes to ensure the accuracy and consistency of data across multiple systems. For ingestion, we leveraged Kafka and Confluent Kafka clusters as our real-time streaming data bus. Data flowed into Kafka topics from various upstream sources
* Spearheaded Data Migration leveraging SQL, SQL Azure, Azure storage, and Azure Data Factory, including SSIS integration And Transformed data into insightful reports within Analytics team fostering informed decision-making.
* Played a crucial role in the migration of existing Hive/SQL queries to Spark Data Frame-based transformations, onto Azure Data Lake Storage (ADLS) Gen2, enhancing data processing capabilities and ensuring seamless integration with modern data architectures.

* Developed optimized solutions with complex transformations on Azure Databricks cluster using Spark(Scala) and PySpark, tested workloads in flink.
* Worked on solutions and designed scalable machine learning pipelines leveraging Python, Scikit-learn, and TensorFlow, enhancing dataset performance and enabling real-time analysis, as well as facilitating the fine-tuning of predictive models for improved accuracy and insights.
* Automated scheduled data purge processes via Azure Data Lake, Azure Data Factory, and Python, achieving a reduction of storage costs.
* Designed and modified tables, stored procedures, and views within the Azure SQL database, enabling the population of Fact and Dimension tables that serve as the foundation for powerful data visualization
* Designed Power BI dashboard reports aligned with business requirements, enhancing data visualization. Optimized data collection, analysis, and visualization methods for improved efficiency. Collaborated with data scientists and analysts to understand data requirements and provide optimal data solutions that meet business objectives.
* Worked closely with stakeholders to define data governance policies and ensure compliance with privacy regulations.
* Conducted performance tuning and optimization of database queries, enhancing overall system performance.
* Collaborated with cross-functional teams to integrate new data sources into existing data pipelines, ensuring seamless data flow.
* Maintained documentation for data pipelines, data models, and data governance processes, facilitating knowledge transfer and onboarding.
* For ingestion, we leveraged Kafka and Confluent Kafka clusters as our real-time streaming data bus. Data flowed into Kafka topics from various upstream sources both within internal and external partner systems.
* To enable data transformation and routing, we used KSQL for filtering ,processing and deriving new data streams in Kafka. KSQL provided SQL-like functionality without needing separate processing clusters.
* I worked on building scalable and fault-tolerant data pipelines to powering real-time payment processing capabilities.

Quality Analyst | Amazon | India **FEB 2017 - JUL 2018**

Implemented security best practices and compliance measures in accordance with AWS Well- Architected Framework.

* Implemented Infrastructure as Code using AWS CloudFormation, automating the provisioning and management of AWS resources.
* Designed and implemented serverless applications, leveraging AWS Lambda, API Gateway, and DynamoDB.
* Spearheaded the adoption of serverless architecture using AWS Lambda, reducing development time and infrastructure costs by 25%.
* Conducted training sessions for development teams to promote serverless best practices and coding standards.
* Led the end-to-end migration of on-premises infrastructure to AWS, resulting in a 40% reduction in operational costs and improved scalability.
* Implemented AWS services such as EC2, S3, and RDS to replicate and enhance on-premises Utilized AWS Migration Hub for tracking and monitoring the progress of the migration. Used to analyze the financial statements of the companies in order to predict the future sales, profit/loss and share values.
* Updating key financial highlights on different & customized financials decks for each type of sector including trading and transaction comps, relative valuation.
* Implemented CI/CD pipelines using AWS Code Pipeline for automated deployment of serverless applications.
* Data cleaning, De-duplication, validation, consolidation by reviewing data reports and indicators, add/edit data as per process guidelines.
* Conducting research to gather critical business information through creatively using sourcing techniques for internal & external databases such as Contributor reports, Annual Reports, SEC/SEDAR filings etc. and conducting qualitative analysis.
* Developed scripts (Python) to Extract, Load, and Transform data. Used the Waterfall methodology to build the different phases of the Software Development Life Cycle. Involve in requirements gathering, analysis, design, development, testing production of an application using the Agile model. Built and maintained data warehouse solutions using technologies like Snowflake, optimizing data storage and retrieval for analytical purposes.
* Conducted data modeling and schema design to ensure effective organization and storage of data in databases.
* Designing and deploying dashboards with Tableau and providing complex reports, including charts, summaries, and graphs to interpret the findings to the team and stakeholders.
* Initiatives for reporting, web review, benchmarking, and content development using Google Analytics.
* Data extraction from various flat files, MS Excel, MS Access, and transformation of the data depending on user requirements and loading data into the target by scheduling sessions.
* Used Python, implemented multiple machine learning techniques such as Decision Tree, Naive Bayes, Logistic Regression, and Linear Regression to evaluate the accuracy rate of each model.
* Stayed up to date with the latest cloud technologies, data analytics tools, and industry trends, and recommended innovative solutions to enhance data analytics capabilities.

EDUCATION

Master's in Industrial Engineering, Central Michigan University 08/2019 - 05/2021| Mt Pleasant, MI

Bachelor's in Industrial Engineering, Gitam University 08/2012 - 05/2016 | Vizag, India